

Lists

```
movies = ["Leo", "Vikram", "Master", "Jailer", "KGF"]
print(movies[0])
print(movies[-1])
```

```
Leo
KGF
```

```
movies = ["Leo", "Vikram", "Master", "Jailer", "KGF"]
movies.append("RRR")
movies.pop(1)
print(movies)
```

```
['Leo', 'Master', 'Jailer', 'KGF', 'RRR']
```

```
movies = ["Leo", "Vikram", "Master", "Jailer", "KGF"]
movies.sort()
print(movies)
```

```
['Jailer', 'KGF', 'Leo', 'Master', 'Vikram']
```

```
movies = ["Leo", "Vikram", "Master", "Jailer", "KGF"]
print(len(movies))
print("Leo" in movies)
```

```
5
True
```

Tuples

```
cities = ("Chennai", "Coimbatore", "Madurai", "Salem", "Trichy")
print(cities[1], cities[3])
```

```
Coimbatore Salem
```

```
cities = ("Chennai", "Coimbatore", "Madurai", "Salem", "Trichy")
# cities[0] = "Delhi"
# Error: Tuples are immutable (cannot be changed)
```

```
cities = ("Chennai", "Coimbatore", "Madurai", "Salem", "Trichy")
a, b, c, d, e = cities
print(a, b, c, d, e)
```

```
Chennai Coimbatore Madurai Salem Trichy
```

```
cities = ("Chennai", "Coimbatore", "Madurai", "Salem", "Trichy")
temp = list(cities)
temp.append("Delhi")
cities = tuple(temp)
print(cities)
```

```
('Chennai', 'Coimbatore', 'Madurai', 'Salem', 'Trichy', 'Delhi')
```

Dictionary

```
student = {"name": "Shobana", "age": 20, "grade": "A", "city": "Chennai"}
print(student)
```

```
{'name': 'Shobana', 'age': 20, 'grade': 'A', 'city': 'Chennai'}
```

```
student = {"name": "Shobana", "age": 20, "grade": "A", "city": "Chennai"}
print(student.keys())
print(student.values())
print(student.items())
```

```
dict_keys(['name', 'age', 'grade', 'city'])
dict_values(['Shobana', 20, 'A', 'Chennai'])
dict_items([('name', 'Shobana'), ('age', 20), ('grade', 'A'), ('city', 'Chennai')])
```

```
student = {"name": "Shobana", "age": 20, "grade": "A", "city": "Chennai"}
student.pop("city")
print(student)
```

```
{'name': 'Shobana', 'age': 20, 'grade': 'A'}
```

Set

```
weekday = {"Maths", "Science", "English"}
weekend = {"Python", "AI", "Maths"}
print(weekday, weekend)
```

```
{'Maths', 'Science', 'English'} {'Maths', 'Python', 'AI'}
```

```
weekday = {"Maths", "Science", "English"}
weekend = {"Python", "AI", "Maths"}
print(weekday | weekend)
print(weekday & weekend)
print(weekday - weekend)
```

```
{'Maths', 'AI', 'Python', 'Science', 'English'}
{'Maths'}
{'Science', 'English'}
```

```
weekday = {"Maths", "Science", "English"}
weekend = {"Python", "AI", "Maths"}
weekday.add("History")
weekend.remove("AI")
print(weekday, weekend)
```

```
{'Maths', 'History', 'Science', 'English'} {'Maths', 'Python'}
```

```
weekday = {"Maths", "Science", "English"}
weekend = {"Python", "AI", "Maths"}
print("Maths" in weekday or "Maths" in weekend)
```

```
True
```

Functions

```
def greet(name):
    print(f"Hello, {name}! Welcome to Day 2.")

greet("Shobana")
```

```
Hello, Shobana! Welcome to Day 2.
```

```
def add_numbers(a, b):
    return a + b

print(add_numbers(5, 10))
```

```
15
```

```
def find_max(a, b, c):
    return max(a, b, c)

print(find_max(3, 7, 5))
```

```
7
```

```
def is_even(n):
    return n % 2 == 0

print(is_even(4))
```

```
True
```

Default Parameters

```
def introduce(name, language="Python"):
    print(f"{name} is learning {language}")
```

```
introduce("Shobana")
```

Shobana is learning Python

```
def introduce(name, language="Python"):
    print(f"{name} is learning {language}")
```

```
introduce("Shobana", "SQL")
```

Shobana is learning SQL

```
def calculate_discount(price, discount=10):
    return price - (price * discount / 100)
```

```
print(calculate_discount(1000))
```

900.0

If-else

```
num = int(input("Enter number: "))
if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

Enter number: 10
Positive

```
score = int(input("Enter score: "))
if score >= 90:
    print("A")
elif score >= 80:
    print("B")
elif score >= 70:
    print("C")
elif score >= 60:
    print("D")
else:
    print("F")
```

Enter score: 100
A

```
year = int(input("Enter year: "))
if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
else:
    print("Not Leap Year")
```

Enter year: 2026
Not Leap Year

For loop

```
for i in range(1, 21):
    print(i)
```

1
2
3
4
5
6
7
8
9
10
11

```
12
13
14
15
16
17
18
19
20
```

```
for i in range(2, 51, 2):
    print(i)
```

```
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
```

```
fruits = ["Apple", "Banana", "Mango"]
for i, fruit in enumerate(fruits):
    print(i, fruit)
```

```
0 Apple
1 Banana
2 Mango
```

```
total = 0
for i in range(1, 101):
    total += i
print(total)
```

```
5050
```

```
text = "hello world"
vowels = "aeiou"
count = 0

for ch in text:
    if ch in vowels:
        count += 1

print(count)
```

```
3
```

Numpy

```
import numpy as np
```

```
arr = np.array([10, 20, 30, 40, 50])
print(arr)
```

```
[10 20 30 40 50]
```

```
print(np.zeros(10))
print(np.ones(10))
```

```
[0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
[1. 1. 1. 1. 1. 1. 1. 1. 1. 1.]
```

```
print(np.arange(1,21))
```

```
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]
```

```
print(np.linspace(0,1,7))
```

```
[0.          0.16666667 0.33333333 0.5          0.66666667 0.83333333
 1.          ]
```

```
print(np.random.randint(1,101,8))
```

```
[85 49 45 71 62 98 55 76]
```

```
print(np.zeros((3,3)))
```

```
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
```

```
print(np.eye(3))
```

```
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

```
print(np.full((4,4),7))
```

```
[[7 7 7 7]
 [7 7 7 7]
 [7 7 7 7]
 [7 7 7 7]]
```

```
print(np.random.rand(3,4))
```

```
[[0.43923094 0.49447735 0.76580945 0.48531418]
 [0.21667944 0.68609593 0.74803324 0.09604469]
 [0.6248683  0.55238684 0.27912878 0.63100391]]
```

```
arr = np.array([[1,2,3],[4,5,6],[7,8,9]])
print(arr)
```

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

Start coding or [generate](#) with AI.